








Blockchain Applications

Lecture 4: Volatility and Tail Risk



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Opening: One-Day Loss Risk of a Crypto Position



Robert F. Engle. **Photo: Jon Demske (2022); CC BY-SA 4.0; Wikimedia Commons**

"These are mean zero, serially uncorrelated processes with nonconstant variances conditional on the past, but constant unconditional variances. For such processes, the recent past gives information about the one-period forecast variance."

— *Engle (1982)*

- GARCH(1,1)-t, the generalised autoregressive conditional heteroskedasticity model with Student-t errors, 2018–2026
  - for crypto, $\hat{\alpha} + \hat{\beta} = 1$: in the model, shocks never fade; $\hat{\nu} \approx 3.1$: very heavy tails
- Bitcoin's annualised conditional volatility: 253.5% on 13 March 2020, 44.6% on 18 September 2026
- Bitcoin one-day value at risk at 1% (VaR 1%) on 19 September 2026: 5.4% to 7.8% of the position, depending on the method
  - expected shortfall at 2.5% (ES 2.5%): up to 8.0%
- Backtest 2020–2026: the Normal and the exponentially weighted moving average (EWMA) models fail the Kupiec test
  - 45 and 51 violations, where 24.5 are expected

▸ Central chart: Conditional Volatility

Lecture Question and Outline

- **Question**: how much can a crypto position lose tomorrow, and how do we know the risk model is right?
  - an exchange sets margins, a fund reports risk, a firm with crypto on its balance sheet needs a loss budget
- **Outline**
  - from volatility clustering (Lecture 2) to volatility models: exponentially weighted moving average (EWMA), generalised autoregressive conditional heteroskedasticity (GARCH)
  - estimation by maximum likelihood with Student-t errors; diagnostics
  - Value-at-Risk (VaR) and Expected Shortfall (ES): four methods
  - backtesting: Kupiec, Christoffersen, the Basel traffic light, an ES check
  - out of sample: the forecasts meet new data
- Data: daily log returns in %, 1 January 2018 – 18 September 2026 (EODHD)

Learning Outcomes

- Write down EWMA and GARCH(1,1) and interpret ω, α, β, persistence and half-life
- Read a GARCH estimation table with Student-t errors and judge the model with residual diagnostics
- Define VaR and ES, and compare Normal, historical, EWMA and GARCH-t estimates
- Interpret the Kupiec, Christoffersen and conditional-coverage tests and the Basel traffic light
- Judge what a short out-of-sample period can and cannot show

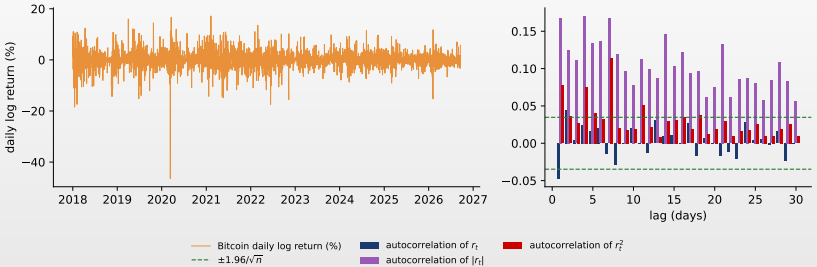
Abbreviations

AIC Akaike information criterion
ATS Advanced Time Series Analysis and Forecasting (course)
BTC Bitcoin
CDF cumulative distribution function
DM Diebold–Mariano (test)
EODHD EOD Historical Data (data vendor)
ES expected shortfall
ETF exchange-traded fund
EWMA exponentially weighted moving average
FABIZ Faculty of Business Administration in Foreign Languages, ASE
GARCH generalised autoregressive conditional heteroskedasticity
GARCH-t GARCH with Student-t errors
GJR Glosten–Jagannathan–Runkle (asymmetric GARCH)
HAC heteroskedasticity- and autocorrelation-consistent
HS historical simulation
IGARCH integrated GARCH
i.i.d. independent and identically distributed
KS Kolmogorov–Smirnov (test)
LR likelihood ratio
MAR market risk standard of the Basel Framework
MSc Master of Science
MV minimum variance
PIT probability integral transform
QLIKE quasi-likelihood loss
RC risk contribution
RMS root mean square
S&P 500 Standard & Poor's 500 stock index
sd standard deviation
se standard error
TLT iShares 20+ Year Treasury Bond ETF
US United States
USD US dollar
VaR value-at-risk

Reading and Tools

- Volatility models: Engle (1982), Bollerslev (1986), Glosten, Jagannathan & Runkle (1993); for crypto: Katsiampa (2017), Chu et al. (2017)
- Risk measures and backtests: Artzner et al. (1999), Kupiec (1995), Christoffersen (1998), McNeil & Frey (2000), Gneiting (2011)
- Regulation: Basel Committee (2019), market-risk standard; Basel Committee, MAR33 (ES), Basel Committee, MAR32 (backtesting)
- Forecast evaluation: Patton (2011), Diebold & Mariano (1995), Berkowitz (2001)
- Video unit on Quantinar: Robust Markowitz for Crypto-Assets
- Related lectures: Advanced Time Series Analysis and Forecasting (ATS), *GARCH Models* and *VaR & ES* (2025); MSc dissertation A.-T. Ginavar, *Risk models for digital assets* (VaR/ES backtests for six cryptos)
- Estimation: Python, package `arch`

Volatility Clustering in Bitcoin Returns



- Left: Bitcoin daily log returns, 2018 – 18 September 2026; calm and turbulent periods alternate
- Right: autocorrelations at lags 1–30 days; dashed: $\pm 1.96/\sqrt{n}$, the 5% band under independence
- Returns r_t: 2 of 30 lags outside the band; absolute returns $|r_t|$: all 30 above it (0.06 to 0.17)
- Squared returns r_t^2: 8 of 30 lags above the band; a few extreme days (12 March 2020: -46%) dominate them

BA_L4_clustering ▶ Interactive

Models of Volatility Clustering

- The chart (and Lecture 2): returns are hardly autocorrelated, absolute and squared returns are autocorrelated
- Model the return as a mean plus a shock with time-varying scale:

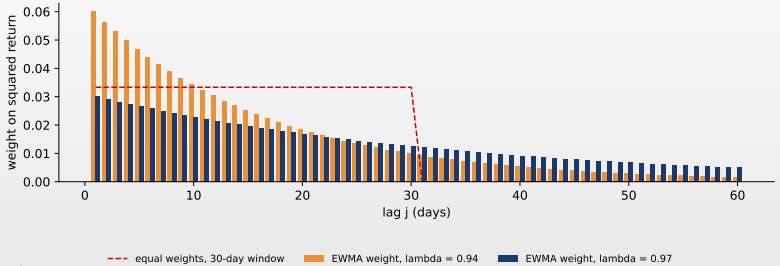
$$r_t = \mu + \varepsilon_t, \qquad \varepsilon_t = \sigma_t z_t, \qquad z_t \overset{\text{i.i.d.}}{\sim} (0,1)$$

  - $\sigma_t^2 = \mathrm{Var}(r_t \mid \mathcal{F}_{t-1})$: the **conditional variance**, known at the end of day $t-1$
  - $\mathcal{F}_{t-1}$: the information up to day $t-1$
- **EWMA** (RiskMetrics): an exponentially weighted average of past squared shocks

$$\sigma_t^2 = \lambda\,\sigma_{t-1}^2 + (1-\lambda)\,\varepsilon_{t-1}^2 = (1-\lambda)\sum_{j\geq 1}\lambda^{j-1}\varepsilon_{t-j}^2, \qquad \lambda = 0.94$$

  - weights decay geometrically; effective memory $1/(1-\lambda) \approx 17$ days
  - no long-run level: after a shock, volatility never returns to a mean

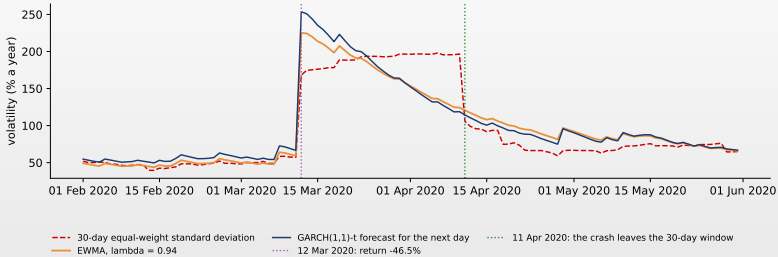
How EWMA Weights the Past



- Weight $(1-\lambda)\lambda^{j-1}$ on the squared return j days ago; red: a 30-day equal-weight window
- $\lambda = 0.94$: 6% on yesterday, 1.0% on day 30 (0.06×0.94^{29}); the first 30 days carry $1 - 0.94^{30} = 84\%$ of the weight
- No cliff at the end of the window: a crash 31 days ago does not suddenly drop out, as it does with a rolling standard deviation

BA_L4_ewma_weights ▶ Interactive

EWMA against a 30-Day Window: Bitcoin around 12 March 2020



- Estimates available at the end of each day, annualised with $\sqrt{365}$: 30-day equal-weight standard deviation, EWMA ($\lambda = 0.94$), GARCH(1,1)-t forecast for the next day
- 12 March 2020 (return -46.5%): EWMA jumps to 224.9% a year, GARCH-t to 253.5%; the 30-day standard deviation only to 168.3%
- The 30-day standard deviation peaks late, at 198.0% on 6 April, 25 days after the crash
- 11 April 2020, the crash leaves the window: the 30-day standard deviation falls from 196.7% to 105.6% in one day; EWMA from 124.3% to 120.6%

BA_L4_window_crash

Worked Example: EWMA on 12 March 2020

- Inputs
  - EWMA variance at the end of 11 March 2020: $\hat{\sigma}^2 = 9.615$, i.e. 3.101% a day, 59.2% a year
  - Bitcoin return on 12 March 2020: $r = -46.47\%$, so $r^2 = 2\,159.7$
- Update with $\lambda = 0.94$

$$\hat{\sigma}^2_{12\ \text{Mar}} = 0.94 \times 9.615 + 0.06 \times 2\,159.7 = 9.038 + 129.58 = 138.62$$

- Daily volatility $\sqrt{138.62} = 11.77\%$; per year $11.77 \times \sqrt{365} = 224.9\%$
- One day supplies 129.58 of the 138.62, i.e. 93% of the new variance
- The crash term shrinks by the factor 0.94 every day
  - 30 days later, on 11 April: $129.58 \times 0.94^{30} = 20.2$, still in the estimate
  - a 30-day window drops the whole crash term on that day

GARCH(1,1)

- Bollerslev (1986):

$$\sigma_t^2 = \omega + \alpha\,\varepsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2, \qquad \omega > 0,\ \alpha, \beta \geq 0$$

  - α: reaction to yesterday's news; β: memory of yesterday's variance; $\alpha + \beta$: **persistence**
- If $\alpha + \beta < 1$: covariance-stationary, with long-run (unconditional) variance

$$\bar{\sigma}^2 = \frac{\omega}{1 - \alpha - \beta}, \qquad \mathrm{E}_t[\sigma_{t+h}^2] = \bar{\sigma}^2 + (\alpha + \beta)^{h-1}\left(\sigma_{t+1}^2 - \bar{\sigma}^2\right)$$

- **Half-life** of a volatility shock: $h_{1/2} = \ln 0.5 / \ln(\alpha + \beta)$ days
- $\alpha + \beta = 1$: **integrated** GARCH (IGARCH); shocks never die out; no finite long-run variance
  - with $\omega > 0$: $\mathrm{E}_t[\sigma_{t+h}^2] = \sigma_{t+1}^2 + (h-1)\,\omega$, the forecast rises linearly with h
  - EWMA is the case $\omega = 0$, $\alpha = 1 - \lambda$, $\beta = \lambda$: a flat forecast, $\mathrm{E}_t[\sigma_{t+h}^2] = \sigma_{t+1}^2$

Student-t Errors and Maximum Likelihood

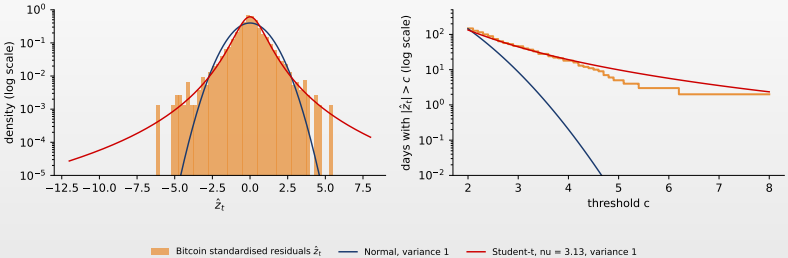
- Heavy tails remain after GARCH filtering: let z_t follow a Student-t with $\nu > 2$ degrees of freedom, scaled to variance 1 (**GARCH-t**)
- Parameters $\theta = (\mu, \omega, \alpha, \beta, \nu)$ by maximum likelihood
  - σ_t^2 computed recursively from θ; $\hat{\theta}$ maximises the Student-t log-likelihood (formula in the Appendix)
  - standard errors: robust (sandwich) covariance of $\hat{\theta}$
- ν measures the tail thickness
  - $\nu \to \infty$: the Normal; ν close to 2: very heavy tails
  - moments of order k exist only for $k < \nu$: with $\nu \leq 4$, $\mathrm{E}[z_t^4]$ is infinite, which matters for the diagnostics

GARCH(1,1)-t Estimates, 2018–2026

	n	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\alpha}+\hat{\beta}$	half-life (days)	$\hat{\nu}$
Bitcoin	3 183	0.144 (0.058)	0.076 (0.012)	0.924 (0.012)	1^*	∞	3.13 (0.15)
Ethereum	3 183	0.550 (0.373)	0.117 (0.038)	0.883 (0.042)	1^*	∞	3.19 (0.20)
S&P 500	2 190	0.035 (0.009)	0.162 (0.022)	0.819 (0.021)	0.981	36	5.85 (0.71)
Gold	2 269	0.015 (0.008)	0.061 (0.018)	0.931 (0.021)	0.992	90	4.06 (0.38)

- Daily log returns in %; constant mean; robust standard errors in brackets; * at the bound $\alpha+\beta\le 1$ imposed in estimation
- Interpretation
  - crypto: $\hat{\alpha}+\hat{\beta}=1$ with $\hat{\omega}>0$, an IGARCH; no finite long-run variance, no mean reversion of volatility in this sample
  - S&P 500: a shock to volatility halves in 36 trading days; long-run volatility $\sqrt{252\,\hat{\omega}/(1-\hat{\alpha}-\hat{\beta})}=21.6\%$
  - $\hat{\nu}\approx 3.1$ for crypto, 5.9 for the S&P 500: crypto shocks are heavy-tailed even after filtering

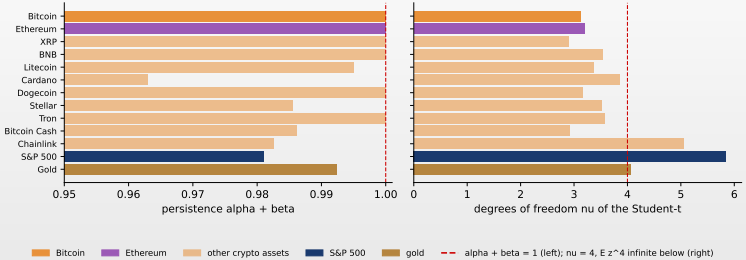
Heavy Tails after Filtering: Bitcoin's Standardised Residuals



- Left: histogram of $\hat{z}_t = (r_t - \hat{\mu})/\hat{\sigma}_t$ on a log scale, against the Normal and the fitted Student-t ($\hat{\nu} = 3.13$), both with variance 1
- Right: number of days with $|\hat{z}_t| > c$, observed (orange) and expected, $n\,P(|z_t| > c)$ with $n = 3\,183$
- $|\hat{z}_t| > 4$: 18 days observed; the Normal expects 0.2, the Student-t 19.2
- $|\hat{z}_t| > 6$: 3 days observed, among them 12 March 2020 with $\hat{z}_t = -13.3$; the Normal expects 6×10^{-9}, the Student-t 5.7

BA_L4_tails ▶ Interactive

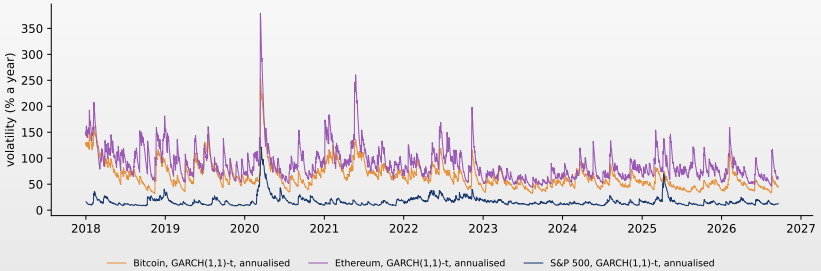
Thirteen Assets: Persistence and Tail Thickness



- GARCH(1,1)-t for 11 crypto assets, the S&P 500 and gold: same model and sample as the table, 2018 – 18 September 2026
- Left: 6 of the 11 crypto assets reach the bound $\hat{\alpha} + \hat{\beta} = 1$ (Bitcoin, Ethereum, XRP, BNB, Dogecoin, Tron); the lowest persistence is Cardano's, 0.963
- Right: $\hat{\nu}$ ranges from 2.90 (XRP) to 5.05 (Chainlink); 10 of the 11 crypto assets have $\hat{\nu} < 4$, an infinite $\mathrm{E}[z_t^4]$
- S&P 500: persistence 0.981, $\hat{\nu} = 5.85$; gold: 0.992 and 4.06

BA_L4_cross_coin

Conditional Volatility



- $\hat{\sigma}_t\sqrt{A}$ from GARCH(1,1)-t, $A = 365$ for crypto, 252 for the S&P 500
- Peaks: Bitcoin 253.5% (13 March 2020), Ethereum 378.4%, S&P 500 121.4% (17 March 2020)
- 18 September 2026: Bitcoin 44.6%, Ethereum 61.2%, S&P 500 12.7%; Bitcoin median 59.2%

BA_L4_condvol ▶ Interactive

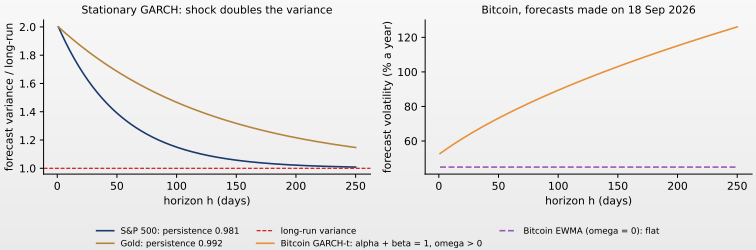
Worked Example: One GARCH Step, 18 September 2026

- Bitcoin estimates: $\hat{\omega} = 0.144$, $\hat{\alpha} = 0.076$, $\hat{\beta} = 0.924$, $\hat{\mu} = 0.056$
- Day T = 18 September 2026
  - conditional variance $\hat{\sigma}_T^2 = 5.453$, i.e. 2.335% a day, 44.6% a year
  - return $r_T = 5.720\%$, shock $\hat{\varepsilon}_T = 5.720 - 0.056 = 5.664$, so $\hat{\varepsilon}_T^2 = 32.08$
- Forecast for 19 September:

$$\hat{\sigma}_{T+1}^2 = 0.144 + 0.076 \times 32.08 + 0.924 \times 5.453 = 0.144 + 2.440 + 5.038 = 7.623$$

- $\hat{\sigma}_{T+1} = \sqrt{7.623} = 2.761\%$ a day, $2.761 \times \sqrt{365} = 52.7\%$ a year: the input of the VaR worked example
- A rise of 5.7% raises the variance as much as a fall of 5.7%: GARCH(1,1) reacts to ε^2, not to its sign
- Two days ahead, with $\hat{\alpha} + \hat{\beta} = 1$: $\mathrm{E}_T[\sigma_{T+2}^2] = 7.623 + 0.144 = 7.767$, i.e. 53.2% a year

Volatility Forecasts by Horizon after a Shock



- Left, $\hat{\alpha} + \hat{\beta} < 1$: forecast variance relative to the long-run level, after a shock that doubles the variance
  - S&P 500 (0.981): half-life $\ln 0.5 / \ln 0.981 = 36.1$ days; after 250 days the shock is gone
  - gold (0.992): half-life 90 days; still 15% above the long run after 250 days
- Right, Bitcoin, forecasts made on 18 September 2026: no long-run level to return to
  - GARCH-t ($\hat{\alpha} + \hat{\beta} = 1$, $\hat{\omega} = 0.144$): the variance grows by $\hat{\omega}$ a day; volatility 52.7% a year at $h = 1$, 126% at $h = 250$
  - EWMA ($\omega = 0$): flat at 44.9%; only here does $\sqrt{h}$ scaling of today's volatility hold exactly

BA_L4_term_structure ▶ Interactive

Diagnostics: Remaining Volatility Clustering

- Standardised residuals $\hat{z}_t = (r_t - \hat{\mu})/\hat{\sigma}_t$ should be i.i.d.: Ljung–Box $Q(10)$ of $\hat{z}_t^2$ against χ^2_{10} (5% critical value 18.31)

	$Q(10)$ of r_t^2	$Q(10)$ of $\hat{z}_t^2$	p-value for $\hat{z}_t^2$ (χ^2_{10})
Bitcoin	97.6	5.5	0.855
Ethereum	131.8	8.8	0.552
S&P 500	2 338.8	12.0	0.283
Gold	434.4	27.9	0.002

- Interpretation
  - crypto and the S&P 500: we do not detect autocorrelation in $\hat{z}_t^2$ at lags 1–10; this does not prove that all dependence is gone
  - gold: some dependence remains (p 0.002); a richer model (asymmetry, regime changes in 2025–2026) is needed
  - caveat: the χ^2_{10} reference needs $\mathrm{E}[z_t^4] < \infty$; with $\hat{\nu} \approx 3$ it is infinite (next slide)
  - $Q(10)$ grows with the sample size: Lecture 2's 212.7 is for the longer sample since 2014, so the two values are not comparable

Diagnostics with Heavy Tails

- Two checks that do not need $\mathrm{E}[z_t^4] < \infty$
  - permutation p-value: reshuffle $\hat{z}_t$ 999 times; share of reshuffled $Q(10)$ of $\hat{z}_t^2$ at least as large as the observed one
  - Ljung–Box $Q(10)$ of $|\hat{z}_t|$: needs only a finite variance of $|z_t|$, i.e. $\nu > 2$

	$Q(10)$ of $\hat{z}_t^2$	permutation p	$Q(10)$ of $\lvert\hat{z}_t\rvert$	p-value (χ^2_{10})
Bitcoin	5.5	0.232	24.5	0.006
Ethereum	8.8	0.291	10.5	0.395
S&P 500	12.0	0.244	8.3	0.597
Gold	27.9	0.012	41.8	< 0.001

- Interpretation
  - the χ^2 p-value for Bitcoin's $\hat{z}_t^2$ (0.855) is far from the permutation p-value (0.232): the usual reference is unreliable here
  - Bitcoin: $|\hat{z}_t|$ is still autocorrelated (p 0.006); GARCH(1,1)-t leaves some clustering in the size of the shocks
  - Ethereum and the S&P 500: no remaining dependence detected; gold: rejected by all three checks
  - all p-values treat $\hat{\theta}$ as known; estimation error is ignored

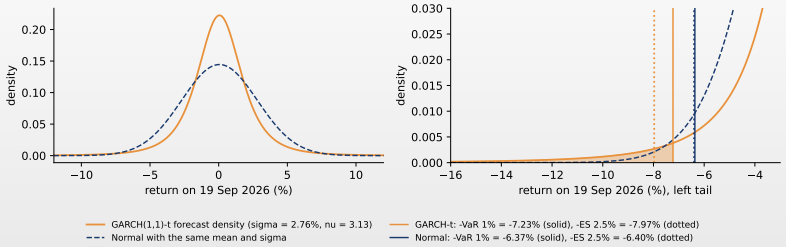
VaR and ES: Definitions

- For a one-day return r_{t+1} with conditional distribution F_t and level p (e.g. 1%):

$$\mathrm{VaR}_p = -F_t^{-1}(p), \qquad \mathrm{ES}_p = -\mathrm{E}\big[r_{t+1} \mid r_{t+1} \le -\mathrm{VaR}_p\big] = \frac{1}{p}\int_0^p \mathrm{VaR}_u \, du$$

  - VaR: the loss exceeded with probability p; ES: the average loss when VaR is exceeded
- Artzner et al. (1999): a **coherent** risk measure is monotone, translation-invariant, positively homogeneous and **subadditive**: $\rho(X+Y) \le \rho(X) + \rho(Y)$
  - ES is coherent; VaR is not subadditive in general, so diversification can appear to raise VaR
  - ES sees the size of the tail; VaR only its threshold
- Regulation (Basel Committee (2019)): for banks using internal models for market-risk capital, ES 2.5% replaced VaR 1% (Basel Committee, MAR33)
  - VaR 1% stays in the backtesting of these models (Basel Committee, MAR32)
  - under the Normal: $\mathrm{VaR}_{1\%} = 2.326\,\sigma$ and $\mathrm{ES}_{2.5\%} = 2.338\,\sigma$, almost equal; under heavy tails ES is larger

VaR and ES on the Forecast Density



- Bitcoin, forecast for 19 September 2026: GARCH(1,1)-t density ($\hat{\sigma}_{t+\mathbf{1}} = 2.76\%$, $\hat{\nu} = 3.13$; method on the next slides) and a Normal with the same mean and standard deviation
- Same variance, different shape: the Student-t is more peaked in the centre and puts more mass far in the tails
- Right, the left tail: VaR 1% is 7.23% under the Student-t and 6.37% under the Normal; ES 2.5% is 7.97% and 6.40%
- Under the Student-t, the Normal VaR 1% is exceeded with probability 1.41%, not 1%
- The two VaRs cross at $p = 1.92\%$; at $p = 5\%$ the Normal VaR is the larger one (4.48% against 3.78%)

BA_L4_density_var ▶ Interactive

Four Ways to Compute VaR and ES

- **Normal**: $\mathrm{VaR}_p = -(\hat{\mu} + \hat{\sigma}\,\Phi^{-1}(p))$, $\mathrm{ES}_p = -\hat{\mu} + \hat{\sigma}\,\varphi(\Phi^{-1}(p))/p$
  - Φ, φ: the standard Normal distribution function and density; $\Phi^{-1}(p)$: its p-quantile
- **Historical simulation** (HS): $\mathrm{VaR}_p = -\hat{q}_p$, the empirical p-quantile of the last W returns; ES_p = minus the mean of returns below $\hat{q}_p$
- **EWMA**: Normal formulas with $\hat{\sigma}_{t+1}$ from the EWMA recursion, $\hat{\mu} = 0$
- **GARCH-t**: with $q = t_\nu^{-1}(p)$ and $s = \sqrt{(\nu-2)/\nu}$

$$\mathrm{VaR}_p = -\big(\hat{\mu} + \hat{\sigma}_{t+1}\, s\, q\big), \qquad \mathrm{ES}_p = -\hat{\mu} + \hat{\sigma}_{t+1}\, s\, \frac{f_\nu(q)}{p}\,\frac{\nu + q^2}{\nu - 1}$$

  - t_ν^{-1}, f_ν: quantile function and density of the standard Student-t; s rescales the t to variance 1
- **Horizon**: $\mathrm{VaR}_h \approx \sqrt{h}\,\mathrm{VaR}_1$ only for i.i.d. Normal returns; with GARCH, use the variance forecasts $\mathrm{E}_t[\sigma^2_{t+j}]$

Worked Example: GARCH-t VaR and ES for Bitcoin

- Inputs on 18 September 2026: $\hat{\sigma}_{t+1} = 2.761\%$, $\hat{\mu} = 0.06\%$, $\hat{\nu} = 3.126$
- Scale factor $s = \sqrt{(\nu-2)/\nu} = \sqrt{1.126/3.126} = 0.600$
- VaR 1%: $q = t_{3.126}^{-1}(0.01) = -4.398$

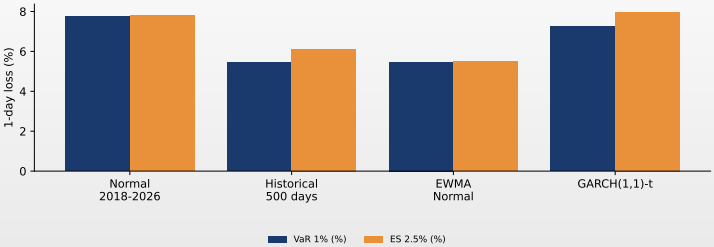
$$\mathrm{VaR}_{1\%} = -0.06 + 2.761 \times 0.600 \times 4.398 = 7.23\%$$

- ES 2.5%: $q = t_{3.126}^{-1}(0.025) = -3.111$; $\dfrac{f_\nu(q)}{p}\dfrac{\nu+q^2}{\nu-1} = 4.844$

$$\mathrm{ES}_{2.5\%} = -0.06 + 2.761 \times 0.600 \times 4.844 = 7.97\%$$

- Under the Normal with the same σ: $2.761 \times 2.338 = 6.45\%$; the t tail adds 1.5 percentage points

VaR and ES by Method



- Bitcoin, one-day horizon, forecasts for 19 September 2026 (table on the next slide)
- Normal and EWMA: ES ≈ VaR; historical and GARCH-t: ES exceeds VaR by 13% and 10%

BA_L4_var_methods ▶ Interactive

Bitcoin: VaR and ES for 19 September 2026

Method	VaR 1% (%)	ES 2.5% (%)	VaR, USD per 1m	ES, USD per 1m
Normal, 2018–2026	7.75	7.79	74 567	74 862
Historical simulation, 500 days	5.43	6.12	52 854	59 002
EWMA, $\lambda = 0.94$, Normal	5.47	5.49	53 208	53 433
GARCH(1,1)-t	7.23	7.97	69 753	75 761

- One-day losses of a long position of 1 million USD, in log-return % and in USD
  - VaR in USD: $10^{6}(1 - e^{-\mathrm{VaR}/100})$, exact, since the USD loss is a monotone function of the return
  - ES in USD: the mean of the USD loss $10^{6}(1 - e^{r/100})$ over the 2.5% tail of r, not $10^{6}(1 - e^{-\mathrm{ES}/100})$
- Interpretation
  - the Normal and EWMA give $\mathrm{ES}_{2.5\%} \approx \mathrm{VaR}_{1\%}$: they ignore the heavy tail
  - GARCH-t: ES exceeds VaR by 10%, the cost of the heavy tail ($\hat{\nu} = 3.13$)
  - the four VaR figures differ by up to 43%: the choice of model is itself a risk (**model risk**)

Backtesting VaR: the Hit Sequence

- Hit (violation): $I_t = \mathbb{1}\{r_t < -\mathrm{VaR}_{p,t}\}$; a correct model has $I_t \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$
- **Unconditional coverage** Kupiec (1995): x hits in n days, $\hat{\pi} = x/n$

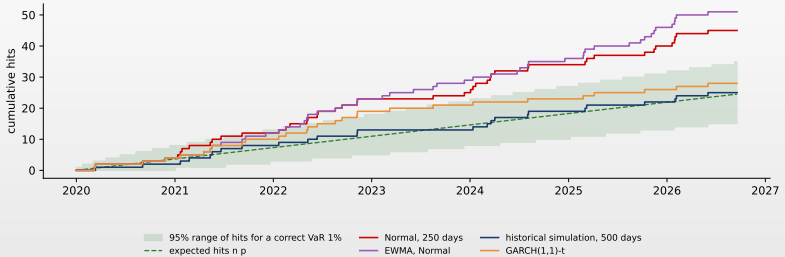
$$\mathrm{LR}_{uc} = -2 \ln \frac{(1-p)^{n-x} p^x}{(1-\hat{\pi})^{n-x} \hat{\pi}^x} \sim \chi^2_1$$

- **Independence** Christoffersen (1998): n_{ij} = number of days in state j after state i; $\hat{\pi}_{ij} = n_{ij}/(n_{i0} + n_{i1})$

$$\mathrm{LR}_{ind} = -2 \ln \frac{(1-\hat{\pi})^{n_{00}+n_{10}} \hat{\pi}^{n_{01}+n_{11}}}{(1-\hat{\pi}_{01})^{n_{00}} \hat{\pi}_{01}^{n_{01}} (1-\hat{\pi}_{11})^{n_{10}} \hat{\pi}_{11}^{n_{11}}} \sim \chi^2_1$$

- **Conditional coverage**: $\mathrm{LR}_{cc} = \mathrm{LR}_{uc} + \mathrm{LR}_{ind} \sim \chi^2_2$
- **Basel traffic light** (Basel Committee, MAR32), 250 days at VaR 1%: green 0–4 hits, amber 5–9 ("yellow" in the 1996 rules), red 10 or more

Cumulative VaR Hits: Four Methods, 2020–2026



- Bitcoin, 1-day VaR 1%: cumulative hits since 1 January 2020 (2 453 days); dashed: the expected *np*; shaded: the 2.5%–97.5% binomial range for a correct VaR, day by day
- End of 2020: 4 hits for Normal-250, EWMA and GARCH-t, 2 for HS; 3.7 expected
- Normal-250 leaves the range in June 2021 and EWMA in April 2022; they end at 45 and 51 hits, above the final range of 15–35
- GARCH-t ends at 28 and HS at 25; GARCH-t lies above the range on 7% of the days, first on 13 September 2022; HS never
- The range holds day by day, it is not a test; the tests on the next slides use the whole sequence

BA_L4_cum_hits

Worked Example: the Kupiec Test

- Normal VaR on a 250-day window, Bitcoin 2020–2026: $n = 2\,453$ days, $x = 45$ hits, $p = 0.01$, $\hat{\pi} = 45/2\,453 = 0.0183$

$$\ln L_0 = 2\,408 \ln 0.99 + 45 \ln 0.01 = -231.4, \qquad \ln L_1 = 2\,408 \ln(1 - 0.0183) + 45 \ln 0.0183 = -224.5$$

$$\mathrm{LR}_{uc} = -2(\ln L_0 - \ln L_1) = 13.84 > 3.84$$

- The model is rejected: 45 hits where 24.5 were expected, almost twice the promised frequency
- For a bank this is a capital issue: Basel multiplies market-risk capital by a factor that rises with the number of exceptions

Worked Example: the Christoffersen Test

- GARCH-t VaR 1%, Bitcoin 2020–2026: 28 hits in 2 453 days, so 2 452 pairs of consecutive days
- Transition counts: $n_{00} = 2\,397$, $n_{01} = 27$, $n_{10} = 27$, $n_{11} = 1$ (one pair of consecutive hits: 8 and 9 November 2022)
- Hit probabilities: after a calm day $\hat{\pi}_{01} = 27/2\,424 = 0.0111$; after a hit $\hat{\pi}_{11} = 1/28 = 0.0357$; overall $\hat{\pi} = 28/2\,452 = 0.0114$

$$\ln L_0 = 2\,424 \ln(1 - 0.0114) + 28 \ln 0.0114 = -153.07$$

$$\ln L_1 = 2\,397 \ln(1 - 0.0111) + 27 \ln 0.0111 + 27 \ln(1 - 0.0357) + \ln 0.0357 = -152.59$$

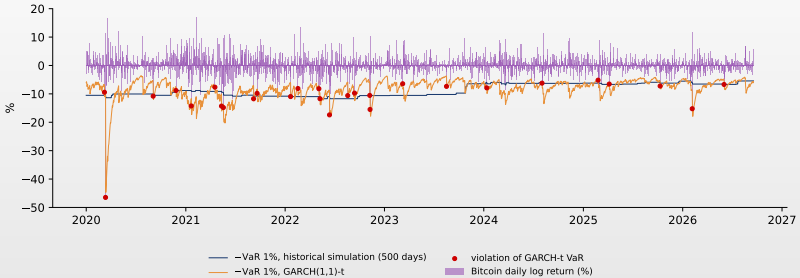
- $\mathrm{LR}_{ind} = -2(\ln L_0 - \ln L_1) = 0.95 < 3.84$ (p 0.329): we do not reject independence
- A hit after a hit is 3.2 times as likely as after a calm day, but one consecutive pair is weak evidence

Backtest: Bitcoin, 1-Day VaR 1%, 2020–2026

Method	hits	rate (%)	LR_{uc} (p)	LR_{ind} (p)	LR_{cc} (p)	last 250	zone	mean VaR (%)
Normal, 250 days	45	1.83	13.84 (<0.001)	0.04 (0.850)	13.88 (0.001)	5	amber	7.16
HS, 500 days	25	1.02	0.01 (0.924)	1.29 (0.256)	1.30 (0.523)	3	green	8.64
EWMA, Normal	51	2.08	22.01 (<0.001)	0.00 (0.952)	22.01 (<0.001)	5	amber	6.78
GARCH(1,1)-t	28	1.14	0.47 (0.491)	0.95 (0.329)	1.43 (0.490)	2	green	8.12

- 2 453 out-of-sample days, 1 January 2020 – 18 September 2026; expected hits $np = 24.5$
- No look-ahead: each VaR uses only data up to the previous day; GARCH-t parameters refreshed every 21 days on an expanding window, the variance updated daily
- Interpretation
  - Normal and EWMA fail unconditional coverage: almost twice as many hits as promised
  - HS and GARCH-t: none of the three tests rejects; GARCH-t reacts to news, HS needs a higher average VaR (8.64% vs 8.12%) for similar coverage

Violations Through Time



- The HS line moves in steps and lags the market; the GARCH-t line widens within days after a shock
- Hits of GARCH-t are spread over time: no clustering, in line with LR_{ind} (p 0.329)

BA_L4_backtest ▶ Interactive

Checking Expected Shortfall

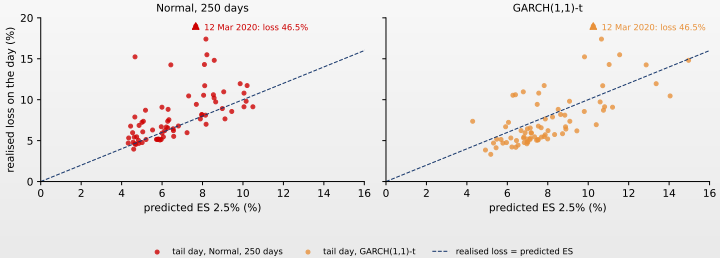
- ES is not elicitable on its own (Gneiting (2011)); a simple check in the spirit of McNeil & Frey (2000): on the days beyond the 2.5% quantile, compare realised loss and predicted ES

$$\bar{R} = \frac{1}{N} \sum_{t:\, r_t < -\mathrm{VaR}_{\mathbf{2.5\%},t}} \frac{-r_t}{\mathrm{ES}_{\mathbf{2.5\%},t}}, \qquad \text{correct model: } \bar{R} \approx 1$$

Method	tail days N	rate (%)	mean loss (%)	mean ES (%)	$\bar{R}$	95% interval for $\bar{R}$
Normal, 250 days	71	2.89	8.48	6.72	1.26	1.13 – 1.44
HS, 500 days	61	2.49	9.05	9.14	1.00	0.89 – 1.15
EWMA, Normal	76	3.10	7.94	6.11	1.31	1.17 – 1.49
GARCH(1,1)-t	76	3.10	7.92	8.04	0.97	0.88 – 1.09

- Interval: bootstrap of the N ratios, tail days treated as independent; expected tail days $0.025 \times 2\,453 = 61.3$
- Normal-based ES: realised tail losses are 26–31% larger than predicted; both intervals lie above 1
- HS and GARCH-t: the intervals contain 1; no evidence against their ES level in this sample
- GARCH-t: 76 tail days against 61.3 expected (Kupiec at 2.5%: $\mathrm{LR}_{uc} = 3.35$, p 0.067)
- $\bar{R}$ is computed only on each method's own tail days, so it mixes errors in VaR 2.5% and in ES: a diagnostic, not a formal ES backtest

Tail Days: Realised Loss against Predicted ES



- One dot per day beyond the method's 2.5% quantile: horizontal, the ES 2.5% predicted the day before; vertical, the realised loss
- Dashed line: realised loss equal to predicted ES; for a correct ES the tail days scatter around it, with mean ratio 1
- Normal-250: 43 of 71 tail days lie above the line (61%); mean ratio $\bar{R} = 1.26$
- GARCH-t: 18 of 76 above the line (24%), median ratio 0.83; a few large losses lift the mean to $\bar{R} = 0.97$
- 12 March 2020, loss 46.5% (off the scale): predicted ES 7.7% (Normal-250) and 10.2% (GARCH-t)

BA_L4_es_scatter

Out-of-Sample Evaluation of the Forecasts

- Out-of-sample test: estimate on data to day T = 18 September 2026, then compare the forecasts with the next five weeks, to 23 October 2026
- GARCH-t parameters are **frozen** on 18 September; only the filter is updated with new returns

$$\hat{\sigma}^2_{t|t-1} = \hat{\omega} + \hat{\alpha}\,(r_{t-1} - \hat{\mu})^2 + \hat{\beta}\,\hat{\sigma}^2_{t-1|t-2}, \qquad t = T+1, \ldots, T+H$$

- Two questions
  - Was the *level* of volatility right? realised $\sqrt{A\,\overline{r^2}}$ against the 18 September forecast $\sqrt{A\,\frac{1}{H}\sum_h E_T \sigma^2_{T+h}}$
  - Did the VaR hold? hits against *np*
- 35 calendar days for crypto, 25 trading days for the S&P 500 and gold: a short window, so the question is also *how much can 35 days tell us?*

What Each Method Updates as New Data Arrive

Method	parameters	volatility state	estimation window
GARCH-t	fixed on 18 Sep	updated daily with r_{t-1}	2018 – 18 Sep, fixed
EWMA, $\lambda = 0.94$	λ fixed	updated daily with r_{t-1}	none
Normal, daily VaR	mean and sd updated daily	none	2018 to $t-1$, expanding
Normal, volatility level	fixed on 18 Sep	none	2018 – 18 Sep, fixed
HS, 500 days	none	none	last 500 days to $t-1$, rolling

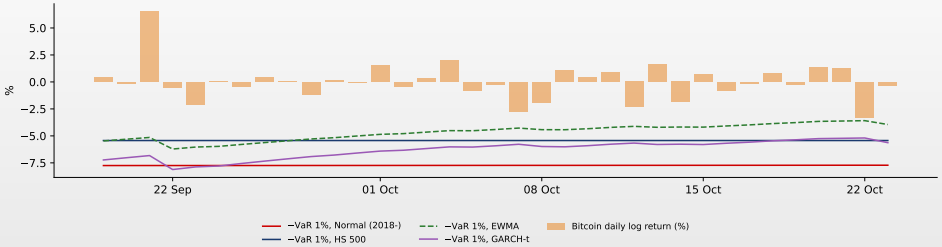
- A forecast can update three things: its parameters, its volatility state, its estimation window
- Fixed-origin forecast (volatility level, next slide): all horizons forecast on 18 September, never updated; it tests the level
- One-step forecast (daily VaR; Appendix: quasi-likelihood (QLIKE) loss, probability integral transform (PIT)): updated as in the table; it tests the reaction to news
- GARCH-t and EWMA react to each new return through their state; the Normal and HS only through their windows

Volatility: Forecast against Realised

annualised, %	Bitcoin	Ethereum	S&P 500	Gold
return over the window (%)	0.2	−9.1	−0.4	−9.3
GARCH-t $\hat{\sigma}_{T+\mathbf{1}}$, 18 Sep	52.7	72.6	11.9	23.3
GARCH-t, average over the window	60.6	93.2	14.4	23.2
EWMA, 18 Sep	44.9	58.8	10.4	23.9
Normal, 2018–2026	64.1	84.3	19.2	16.9
realised, 19 Sep – 23 Oct	31.8	36.1	10.3	19.3
realised / GARCH forecast	0.52	0.39	0.72	0.83

- Every model over-predicted: Bitcoin's realised 31.8% equals the calm of August 2026
- Bitcoin and Ethereum: $\hat{\alpha} + \hat{\beta} = 1$ with $\hat{\omega} > 0$, so the 18 September forecast *rises* linearly in variance; there is no long-run level to revert to
- Bitcoin: from 52.7% for 19 September to 67.6% on day 35 of the window, 60.6% on average
- Gold fell 9.3% with realised volatility 19.3%, below its GARCH forecast (23.2%): a large price move is not the same as high volatility

Bitcoin VaR, Rolled Forward



1-day VaR 1%, 35 days	Normal	HS 500	EWMA	GARCH-t
mean VaR (%)	7.73	5.43	4.66	6.28
mean capital per 1m USD	74 428	52 854	45 500	60 870
hits (expected 0.35)	0	0	0	0

- Worst day −3.27% (22 October): 0.63 of that day's GARCH-t VaR; 18 September forecast for 19 September: VaR 7.23%, realised +0.41%

BA_L4_oos_var_btc ▶ Interactive

Statistical Power of a 35-Day Backtest

- With a correct 1% VaR, the number of hits is $X \sim \mathrm{Bin}(35, 0.01)$:

$$P(X = 0) = 0.99^{35} = 0.703, \qquad \mathrm{LR}_{uc}(X = 0) = -2 \cdot 35 \ln 0.99 = 0.70 \; (p = 0.40)$$

- A model with a *true* hit rate of 2% (as Normal-250 in the backtest) still shows zero hits with probability $0.98^{35} = 0.493$
- Power: to detect 2% against 1% with probability 0.8 at the 5% level (exact one-sided binomial test) one needs about **900 days** (905)
- Conclusions
  - Zero hits in five weeks is *no evidence* for any of the four methods: all four "pass"
  - The methods differ in capital: GARCH-t asks for 35% more than EWMA in this window; the long backtest, not this window, says which one is right
  - Use the whole density, not only the hits (Appendix)

Summary

- GARCH(1,1)-t fits crypto with $\hat{\alpha} + \hat{\beta} = 1$ (IGARCH) and $\hat{\nu} \approx 3.1$: shocks are very heavy-tailed even after filtering
- Diagnostics: no autocorrelation detected in Bitcoin's $\hat{z}_t^2$; with $\hat{\nu} \approx 3$ the check on $|\hat{z}_t|$ still finds some (p 0.006)
- Bitcoin 1-day VaR 1% on 19 September 2026: 5.4% to 7.8% depending on the method; ES 2.5% up to 8.0%
- Backtest 2020–2026: Normal and EWMA fail Kupiec (45 and 51 hits for 24.5 expected); for HS and GARCH-t no test rejects
- With Normal-based ES, realised tail losses are 26–31% larger than predicted (95% intervals above 1)
- Out of sample (to 23 October 2026) all models over-predicted volatility; 35 days cannot discriminate between VaR methods

Interpretation Exercises

1. Bitcoin: $\hat{\alpha} + \hat{\beta} = 1$ (at the bound); S&P 500: 0.981. What does each imply for a 10-day volatility forecast made after a crash?
2. Ljung–Box for gold: $Q(10)$ of $\hat{z}_t^2 = 27.9$ (p 0.002). Is the model adequate? What would you try next?
3. A risk manager uses EWMA-Normal for Bitcoin and had 5 violations in the last 250 days. Which zone is that, and what do the full-period tests say about the model?
4. The GARCH-t ES for 1 million USD is 75 761 USD and the Normal ES is 74 862 USD, yet the backtest favours GARCH-t clearly. Explain using $\bar{R}$.
5. Out of sample (19 Sep – 23 Oct 2026) no method had a Bitcoin VaR hit. A manager concludes that the Normal model is now fine. Respond with the binomial argument.

- Next: Lecture 5, stablecoins and depeg risk
  - a depeg is described by its size, its duration and its probability
  - a daily VaR of returns gives only a loss threshold for one day: none of the three

Appendix: Further Material

Optional material for self-study.

- The Student-t Log-Likelihood
- Testing for a Leverage Effect
- News Impact Curves
- Diagnostics: Normal against Student-t Errors
- The Traffic Light Through Time
- Euler Risk Contributions
- Worked Example: Euler Contributions with Two Assets
- A 60/40 Portfolio with Bitcoin
- Risk Share against Weight
- Bitcoin and the Efficient Frontier
- One-Step Volatility Forecasts, Day by Day
- Forecast Comparison: QLIKE and the Diebold–Mariano Test
- Density Forecast Evaluation: PIT and the Berkowitz Test
- Portfolio Risk and the Instability of Correlations

The Student-t Log-Likelihood

- Log-likelihood of $\theta = (\mu, \omega, \alpha, \beta, \nu)$, with $\varepsilon_t = r_t - \mu$ and σ_t^2 from the GARCH recursion:

$$\ell(\theta) = \sum_{t=1}^{n} \left[\ln \Gamma\left(\tfrac{\nu+1}{2}\right) - \ln \Gamma\left(\tfrac{\nu}{2}\right) - \tfrac{1}{2} \ln\left(\pi(\nu-2)\sigma_t^2\right) - \tfrac{\nu+1}{2} \ln\left(1 + \frac{\varepsilon_t^2}{(\nu-2)\sigma_t^2}\right) \right]$$

  - $\Gamma(x) = \int_0^\infty u^{x-1} e^{-u}\, du$: the gamma function
  - each term is the log density of a Student-t with ν degrees of freedom, scaled to variance σ_t^2
  - $\hat{\theta} = \arg\max \ell(\theta)$ by numerical optimisation, subject to $\omega > 0$, $\alpha, \beta \geq 0$, $\alpha + \beta \leq 1$, $\nu > 2$

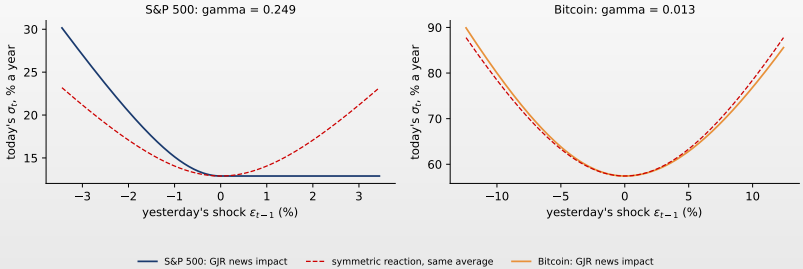
Testing for a Leverage Effect

- GJR-GARCH (Glosten, Jagannathan & Runkle (1993)): $\sigma_t^2 = \omega + \left(\alpha + \gamma\, 1\{\varepsilon_{t-1} < 0\}\right)\varepsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2$
  - $\gamma > 0$: a fall raises volatility more than a rise of the same size; persistence $\alpha + \beta + \gamma/2$
  - test $\gamma = 0$: $\mathrm{LR} = 2(\ell_{\mathbf{GJR}} - \ell_{\mathbf{GARCH}}) \sim \chi_1^2$, 5% critical value 3.84
  - Akaike information criterion $\mathrm{AIC} = 2k - 2\ell$, k = number of parameters: lower AIC is better fit after a penalty for the extra γ

	$\hat{\gamma}$	$t(\hat{\gamma})$	$\hat{\alpha} + \hat{\beta} + \hat{\gamma}/2$	LR	p-value	AIC: GARCH / GJR
Bitcoin	0.013	0.72	1.000	0.68	0.408	15 518 / 15 520
Ethereum	0.028	0.74	0.998	1.19	0.275	17 493 / 17 494
S&P 500	0.249	6.25	0.968	75.89	< 0.001	5 902 / 5 828
Gold	-0.060	-3.21	0.995	14.27	< 0.001	5 980 / 5 968

- S&P 500: falls raise volatility much more than rises; GJR preferred by LR and AIC
- gold: $\hat{\gamma} < 0$: rises raise volatility more than falls
- crypto: no asymmetry detected; AIC prefers GARCH(1,1)

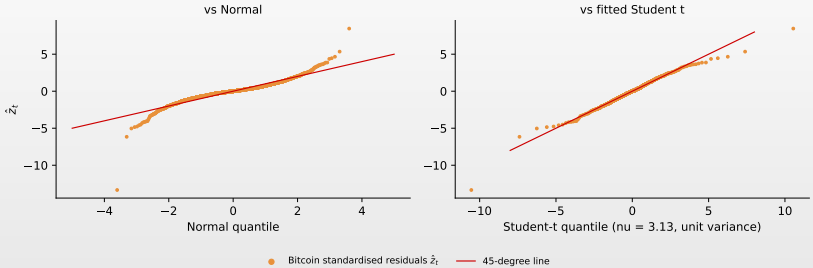
News Impact Curves



- Today's volatility as a function of yesterday's shock, with yesterday's variance at its sample median (GJR-GARCH-t)
- S&P 500: a -3% day raises volatility to 27.0% a year, a $+3\%$ day leaves it at 12.9%: strong asymmetry
- Bitcoin: the curve is almost symmetric ($\hat{\gamma} = 0.013$): good and bad news raise volatility alike

BA_L4_news_impact

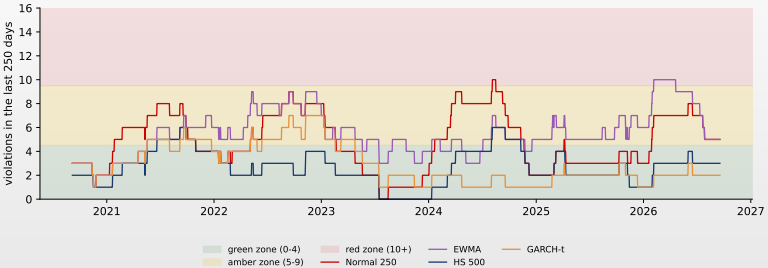
Diagnostics: Normal against Student-t Errors



- Bitcoin $\hat{z}_t$ against Normal quantiles (left) and against the fitted Student-t with $\hat{\nu} = 3.13$ (right)
- The Normal fails in both tails; the Student-t fits well except the single most extreme day (12 March 2020: $\hat{z}_t = -13.3$ against a t quantile of -10.5)

BA_L4_qq_resid

The Traffic Light Through Time



- Number of hits in the last 250 days for each method; background: Basel green, amber and red zones
- Share of days in the amber or red zone: Normal 50.3%, EWMA 75.6% (red 3.4%), historical 10.5%, GARCH-t 22.9% (never red)
- A model can pass a full-sample test and still spend long periods in the amber zone: monitor it continuously

BA_L4_traffic_light

Euler Risk Contributions

- Portfolio return $r_p = w^\top r$, weights w, covariance matrix Σ; volatility $\sigma_p = \sqrt{w^\top \Sigma w}$
- σ_p is homogeneous of degree 1 in w, so by Euler's theorem it is the sum of the risk contributions (RC)

$$\sigma_p = \sum_i w_i \frac{\partial \sigma_p}{\partial w_i} = \sum_i \mathrm{RC}_i, \qquad \mathrm{RC}_i = \frac{w_i (\Sigma w)_i}{\sigma_p}$$

- The same holds for ES; with historical data

$$\mathrm{ES}_p = \sum_i \mathrm{RC}_i^{ES}, \qquad \mathrm{RC}_i^{ES} = -\mathrm{E}\big[w_i r_i \mid r_p \le q_p(r_p)\big]$$

- RC_i/σ_p: the share of risk due to asset i; compare it with the weight w_i
- Minimum-variance (MV) portfolio: $w^{\mathrm{MV}} = \Sigma^{-1}1/(1^\top \Sigma^{-1}1)$

Worked Example: Euler Contributions with Two Assets

- 95% S&P 500 ($\sigma_1 = 19.2\%$), 5% Bitcoin ($\sigma_2 = 63.3\%$), correlation $\rho = 0.29$; daily simple returns 2018–2026, annualised

$$\sigma_p^2 = 0.95^2 \cdot 19.2^2 + 0.05^2 \cdot 63.3^2 + 2 \cdot 0.95 \cdot 0.05 \cdot 0.29 \cdot 19.2 \cdot 63.3 = 376.2, \quad \sigma_p = 19.40\%$$

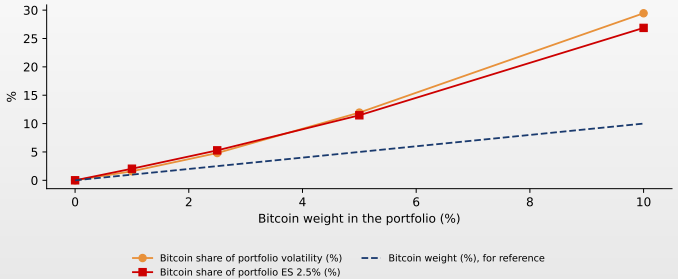
- Contributions $\mathrm{RC}_i = w_i(\Sigma w)_i / \sigma_p$
  - S&P 500: 18.02 percentage points (92.9%); Bitcoin: 1.38 points (7.1%)
  - check: $18.02 + 1.38 = 19.40 = \sigma_p$
- Bitcoin carries 7.1% of the risk with 5% of the money; in the 60/40 portfolio (next slide) it carries 11.9%, because Treasuries diversify the equity risk but not the Bitcoin risk

A 60/40 Portfolio with Bitcoin

Bitcoin weight (%)	volatility (% a year)	ES 2.5% (%, daily)	Bitcoin share of volatility (%)	Bitcoin share of ES (%)
0	12.25	2.26	0.0	0.0
1	12.31	2.29	1.6	2.0
2.5	12.45	2.32	4.8	5.3
5	12.83	2.40	11.9	11.5
10	14.08	2.60	29.4	26.9

- S&P 500 and long US Treasuries in proportion 60/40, plus Bitcoin; 2 190 common days, 2018–2026
  - S&P 500: index GSPC, a price index, so dividends are not included
  - Treasuries: exchange-traded fund (ETF) TLT, adjusted close, so distributions are reinvested
  - daily simple returns; portfolio return $w^\top R_t$, weights rebalanced daily
- Correlations: Bitcoin–S&P 500 0.29, Bitcoin–Treasuries -0.02, S&P 500–Treasuries -0.14
- A 5% Bitcoin weight carries 12% of the portfolio volatility; 10% carries 29%
- Minimum-variance weights: S&P 500 40.5%, Treasuries 59.6%, Bitcoin -0.15%: Bitcoin does not reduce risk

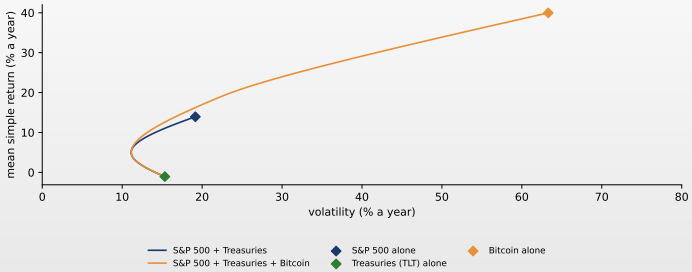
Risk Share against Weight



- The risk share grows faster than the weight: from 1.6 times the weight at 1% to 2.9 times at 10%
- A position limit for crypto should be set on its risk contribution, not on its weight
- Robust alternatives to the sample covariance: Robust Markowitz for Crypto-Assets

BA_L4_riskcontrib

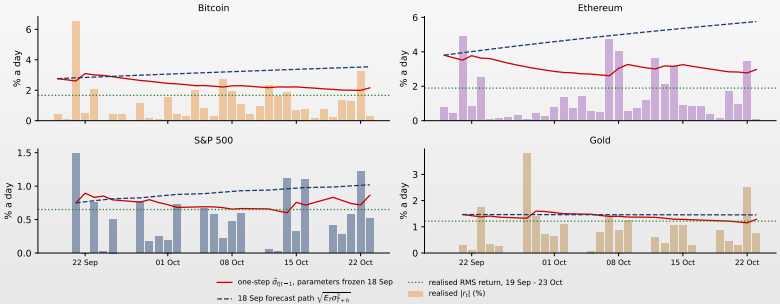
Bitcoin and the Efficient Frontier



- Long-only minimum-variance frontiers, 2018–2026 common days; mean daily simple return $\times 252$ and volatility, both annualised
- Mean return and volatility: Bitcoin 40.0% at 63.3%; S&P 500 13.9% at 19.2% (no dividends); TLT -1.1% at 15.3%
- Compound growth of the net asset value: Bitcoin 21.7%, S&P 500 12.8%, TLT -2.2% a year
- Adding Bitcoin extends the frontier only at high volatility; at the low-risk end the two frontiers coincide: in-sample means, so an upper bound on the benefit

BA_L4_frontier

One-Step Volatility Forecasts, Day by Day



- Red: GARCH-t with frozen parameters reacts to each $|r_t|$ (Bitcoin +6.5% on 21 September); blue: the 18 September term structure
- Ethereum's path goes up: $\hat{\alpha} + \hat{\beta} = 1$ (at the bound) and $\hat{\omega} = 0.55 > 0$, so the forecast variance grows by $\hat{\omega}$ each day

BA_L4_oos_vol

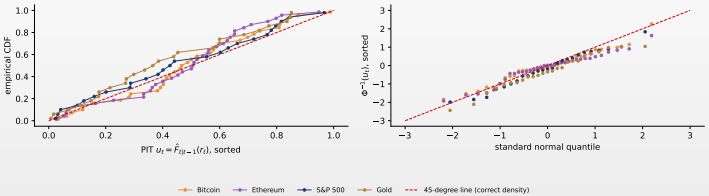
Forecast Comparison: QLIKE and the Diebold–Mariano Test

- Quasi-likelihood (QLIKE) loss, robust to the noise in r_t^2 as a proxy of σ_t^2 (Patton (2011)): $L_t(h) = \ln h_t + r_t^2/h_t$
- Diebold & Mariano (1995): $d_t = L_t(\text{GARCH}) - L_t(\text{other})$, $\mathrm{DM} = \bar{d}/\widehat{\mathrm{se}}_{\mathrm{HAC}}(\bar{d}) \sim N(0,1)$, HAC: heteroskedasticity- and autocorrelation-consistent; $\bar{d} > 0$: GARCH loses

	Bitcoin	Ethereum	S&P 500	Gold
mean QLIKE: GARCH-t	2.232	2.655	0.220	1.480
mean QLIKE: EWMA	2.094	2.435	0.184	1.480
mean QLIKE: constant (Normal)	2.667	3.152	0.670	1.431
GARCH vs EWMA: $\bar{d}$ (DM)	0.138 (1.97)	0.220 (2.99)	0.036 (0.83)	−0.000 (−0.01)
GARCH vs constant: $\bar{d}$ (DM)	−0.435 (−4.11)	−0.497 (−6.12)	−0.450 (−3.50)	0.048 (0.32)

- A time-varying forecast beats the constant one clearly for Bitcoin, Ethereum and the S&P 500
- EWMA beats GARCH for Ethereum ($\mathrm{DM} = 2.99$) and, at the margin, for Bitcoin: it forgets faster, and after a calm month that helps
- But in the 2020–2026 backtest EWMA-Normal failed the VaR test: being good at the *level* is not being good at the *tail*

Density Forecast Evaluation: PIT and the Berkowitz Test



- PIT $u_t = \hat{F}_{t|t-\mathbf{1}}(r_t)$: i.i.d. $U(0,1)$ if the density is right; Berkowitz (2001): $x_t = \Phi^{-\mathbf{1}}(u_t)$, fit $x_t = c + \rho x_{t-\mathbf{1}} + \sigma\varepsilon_t$, $\mathrm{LR} = 2(\ell_{\mathbf{1}} - \ell_{\mathbf{0}}) \sim \chi^2_{\mathbf{3}}$ under $c = 0, \rho = 0, \sigma = 1$

	Bitcoin	Ethereum	S&P 500	Gold
sd of x_t (should be 1)	0.84	0.78	0.98	0.93
Berkowitz LR (p)	3.16 (0.368)	4.80 (0.187)	2.36 (0.501)	5.59 (0.133)
Kolmogorov–Smirnov D (p)	0.125 (0.604)	0.176 (0.203)	0.119 (0.833)	0.186 (0.310)

- sd of x_t below 1 for crypto: predicted densities too wide (over-predicted volatility); not significant in 35 days
- Gold: no test rejects at 5%; mean of x_t is -0.34 (missed downward drift); one 1% hit, -3.78% on Monday 28 September

BA_L4_oos_pit

Portfolio Risk and the Instability of Correlations

60/40 S&P 500/TLT plus Bitcoin Bitcoin weight	volatility (%, ann.) predicted	 realised	Bitcoin share of risk (%) predicted	 realised
0%	12.2	8.1	–	–
5%	12.8	8.7	11.9	13.2
10%	14.1	9.6	29.4	26.6

- Daily simple returns; TLT adjusted close, S&P 500 price index
- Correlation Bitcoin – S&P 500: 0.29 in 2018–2026, 0.76 over the 25 new trading days
- Bitcoin's own volatility halved (63.3% to 31.4% a year), yet its share of risk changed little: the higher correlation offset the lower volatility

$$\text{share}_{\mathbf{BTC}} = \frac{w_B\,(\Sigma w)_B}{w'\Sigma w}, \qquad (\Sigma w)_B = w_B\sigma_B^2 + \textstyle\sum_{j\neq B} w_j\rho_{Bj}\sigma_B\sigma_j$$

- The risk budget is a forecast of the correlation, not only of the volatilities